

Generation Interconnection System Impact Study Report

For Queue Project Q17-0742

Timken Richville 230 kV

November 25, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0742. The Interconnection Customer proposes a 145.5 MW solar photovoltaic facility interconnecting at the Timken Richville 230 kV substation in the Ohio area. The request was received on February 08, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 256 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$9,662,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Cross Plains to N Cambridge 230 kV
Contingency: West Canton to Pt Arena 500 kV
Loading: 126.1% (Rating: 584 MVA)
DFAX: 45.7%
Pre-project loading: 114.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cross Plains to N Cambridge 230 kV	Rebuild line with higher-capacity conductor	\$14,913,000	65% (\$9,662,000)
	Total allocated to project		\$9,662,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.